SYSTEM DESIGN: CAP THEOREM

English lecture notes based on the supplied Hindi YouTube transcript | Approx. 10-minute lesson

1. What is CAP?

CAP stands for **Consistency (C)**, **Availability (A)**, and **Partition Tolerance (P)**. The lecture explains how distributed applications balance these properties.

Consistency: Users reading the same logical data should receive a consistent view. The lecture uses bank balances and movie-seat availability as examples.

Availability: The service should respond to requests. The lecture mentions websites becoming unavailable under heavy demand, such as examination-results sites.

Partition tolerance: A distributed system must deal with communication failures that prevent groups of nodes from exchanging messages.

2. The central CAP trade-off

The lecture presents the common shorthand that a distributed system can choose two of C, A, and P. More precisely, **when a network partition occurs**, a distributed system cannot guarantee both strong consistency and availability for every request. It must prioritize one of them for affected operations.

CP: Prioritize consistent results; some requests may be rejected or delayed during a partition. **AP:** Continue answering requests; some responses may temporarily reflect older data. **CA:** Consistency and availability are possible when no relevant partition is occurring; CA is not a way to tolerate an actual network partition.

3. Examples explained in the lecture

Application	Lecture emphasis	Why
Blog / social feed	Availability	A delayed post update may be acceptable.
Online multiplayer games	Availability	Continuity of service matters to the experience.
Video streaming / chat	Availability	Users expect the service to keep responding.
Banking / balances	Consistency	Incorrect account data can cause serious problems.
Ticket booking	Consistency	Prevent double booking of the same seat.
Stock trading	Consistency	Correctness of orders and account state is crucial.

4. CP scenario: booking the last seat

Suppose two regions cannot communicate. One region confirms the last available seat. To avoid selling the same seat again, the other region can refuse or postpone booking requests until it can verify the latest state. This protects consistency at the expense of availability for those requests.

5. AP scenario: social-media posts

Suppose a new post reaches one replica while another is temporarily disconnected. The disconnected replica may still serve an older feed, keeping the app responsive. When communication recovers, replicas can synchronize. This resembles an eventually consistent approach.

6. Important technical clarifications

Network partition vs server crash: A partition is a communication break between nodes; a server going down is a failure but is not automatically a network partition. Replication can help availability, but replicas alone do not eliminate partition trade-offs.

Availability in CAP: This is a formal response guarantee for non-failing nodes, not simply 24/7 uptime. A CP system can still be highly available in normal operation.

Not every whole application is simply CP or AP: Different features may make different trade-offs. For example, a social app can prioritize feed availability while enforcing stricter consistency for payments.

CA does not mean monolithic: A monolithic application can use distributed databases, and a distributed system can be consistent and available when no partition is present.

7. Interview-ready answer

Question: Explain CAP theorem with an example.
Answer: CAP theorem states that during a network partition, a distributed data system cannot guarantee both consistency and availability. A ticket-booking service may reject bookings to avoid double-selling seats (CP), while a social-media feed may keep serving slightly stale content to remain responsive (AP). The right trade-off depends on the business operation.

8. Quick revision

C: Consistent view of data. **A:** Requests receive responses. **P:** System operates despite communication partitions. **Key rule:** During a partition, choose between preserving consistency and serving every request.

Source note: These notes follow the examples and flow of the supplied lecture transcript. The technical clarifications explicitly refine simplified or inaccurate statements in the transcript.